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**WAGHOLI, PUNE**

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**Department of Information Technology**
**Minor Degree in Cyber Security**
**TAE-2**
**Subject: Fundamental of Cyber Security & Cryptography**

# Mini Project Report On
# "Sentinel: Next-Generation Firewall & Intrusion Detection System"


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**Academic Year 2026-27**
**Semester-V**

# CERTIFICATE

This is to certify that the mini project entitled **“Sentinel: Next-Generation Firewall & Intrusion Detection System”** embodies the original work done by **AMMAR RAISAHMAD MULLA (TY A-16)** as a partial fulfilment of the requirement of TAE-2 for the subject **Fundamental of Cyber Security & Cryptography** under the Minor Degree in Cyber Security, Third Year B. Tech. (Department of Information Technology), Semester V, during the academic year 2026-2027.

**Date:** 01/10/2026
**Place:** Pune

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# 1. Problem Definition & Objectives

## Problem Definition

Modern enterprise computing infrastructure and cloud networks are subjected to persistent, automated cyber-attacks. These range from volumetric Distributed Denial of Service (DDoS) vectors—such as TCP SYN floods designed to saturate server transmission control block (TCB) memory queues—to automated horizontal and vertical port scanning recon probes seeking vulnerabilities across critical administrative daemons (SSH port 22, RDP port 3389, and SMB port 445). Traditional enterprise firewall utilities, including Linux iptables, operate as opaque black boxes providing static text logs that lack real-time interactive telemetry and visual threat trajectory mapping.

## Objectives

- **Layer 4 Stateful Packet Inspection:** To develop a firewall engine evaluating incoming packets against configured rules (IP, port, protocol) with real-time ALLOW/DROP verdicts.
- **Heuristic Intrusion Detection (IDS):** To implement an anomaly engine that flags malicious vectors like TCP SYN Floods (MITRE ATT&CK; T1498.001) and port scans.
- **Interactive Draggable Topology Mesh:** To engineer a dynamic topology graph using SVG math and cubic Bézier curves, providing visual representation of threats and internal LAN assets.
- **Adversary Attack Simulation Console:** To create a simulation console enabling security teams to trigger live attack scenarios on demand to validate packet dropping.
- **Dual Operational Modes:** To support both 'Live Network Mode' (sniffing host TCP sockets via Server-Sent Events) and 'Simulation Mode'.
- **SecOps AI Forensic Analyst:** To integrate an intelligent assistant capable of root-cause analysis on flagged attacker IPs with 1-click drop rules.
- **Relational Database Persistence:** To store firewall rules and logs in MySQL ('firewall_db') with persistent hit and byte counters.

# 2. Introduction & Theoretical Background

The Sentinel Firewall & Intrusion Detection System is an advanced SecOps platform designed to combine stateful packet filtering, heuristic threat signature matching, live socket streaming, and interactive topology modeling. This chapter covers the theoretical principles underpinning next-generation firewall architectures.

## 2.1 Next-Generation Firewalls (NGFW) vs Traditional Packet Filters

Traditional packet filters operate statelessly at Layers 3 and 4, inspecting IP headers in isolation. Next-Generation Firewalls (NGFWs) integrate stateful inspection, connection state tracking (SYN, SYN-ACK, ESTABLISHED), and application awareness. Sentinel combines stateful policy evaluation with heuristic anomaly detection.

## 2.2 Common Network Attack Vectors

- **TCP SYN Flood DDoS (MITRE T1498.001):** Exploits the three-way handshake by transmitting rapid SYN frames without completing handshakes, overflowing kernel connection tables.
- **Port Reconnaissance (MITRE T1046):** Scanning common ports (22, 80, 443, 445, 3389) to enumerate vulnerable services.
- **Credential Brute-Force:** Repeated connection bursts against remote access daemons (SSH/RDP).

**Table 1 : Traditional Firewalls vs Sentinel Next-Gen Firewall**

| Parameter | Traditional Packet Filter | Sentinel Next-Gen Firewall |
|---|---|---|
| Inspection Layer | Layer 3 & 4 only | Layer 3, 4 & Heuristic Signatures |
| Connection State | Stateless | Stateful + Velocity Tracking |
| Intrusion Detection | None (separate daemon) | Built-in Heuristic IDS Engine |
| Attack Simulation | Requires external tools | Integrated Live Simulation Console |
| Topology Modeling | Textual CLI iptables | Interactive Draggable Mesh (Bézier curves) |
| Database Sync | Flat text files | Relational MySQL database |
| AI Forensics | None | Sentinel AI Analyst (Gemini / DeepSeek) |

# 3. System Architecture & Data Pipelines

Sentinel follows a decoupled three-tier architecture ensuring sub-millisecond evaluation latency, browser rendering stability, and persistent database state across sessions.

## 3.1 Architectural Overview

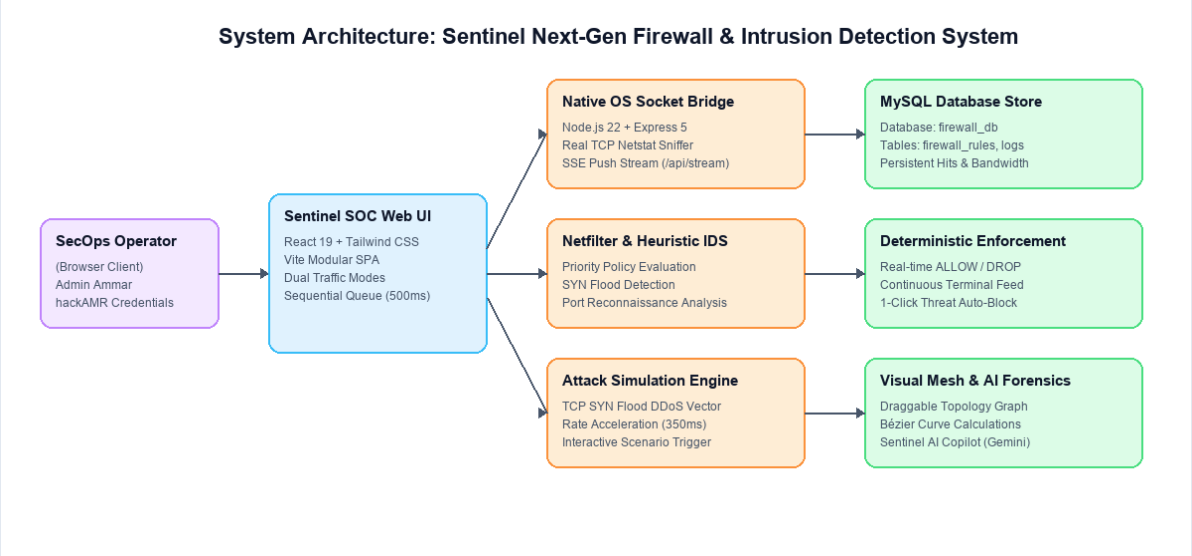



**Fig 1 : System Architecture of Sentinel Next-Gen Firewall & IDS Platform**

## 3.2 Core Functional Modules

**Table 3 : Core System Modules**

| Module | Input | Processing Logic | Output |
|---|---|---|---|
| Live Terminal | Packet stream | Rule matching & formatting | Monospace log stream |
| Netfilter Engine | IP, Port, Protocol | Hierarchical rule evaluation | ALLOW/DROP verdict |
| Heuristic IDS | Live packets | Velocity & signature inspection | Threat feed & alerts |
| Topology Mesh | Origin & LAN nodes | Pointer dragging & Bézier curves | Interactive network mesh |
| Attack Console | Attack vector config | High-rate SYN packet flood | DDoS load & spike |
| AI Analyst | Forensic queries | Live telemetry + Gemini API | Root-cause analysis |

# 4. Implementation: Design Web Pages & UI Modules

The user interface is engineered as a tactical Security Operations Center (SOC) dashboard. The figures below document the implemented views with actual captured system screenshots and reference placeholders.

## 4.1 Administrator Authentication & Access Portal

To guarantee role-based access control (RBAC) and prevent unauthorized modification of netfilter security policies, Sentinel implements a hardened administrative authentication portal requiring credential validation before exposing the tactical dashboard.

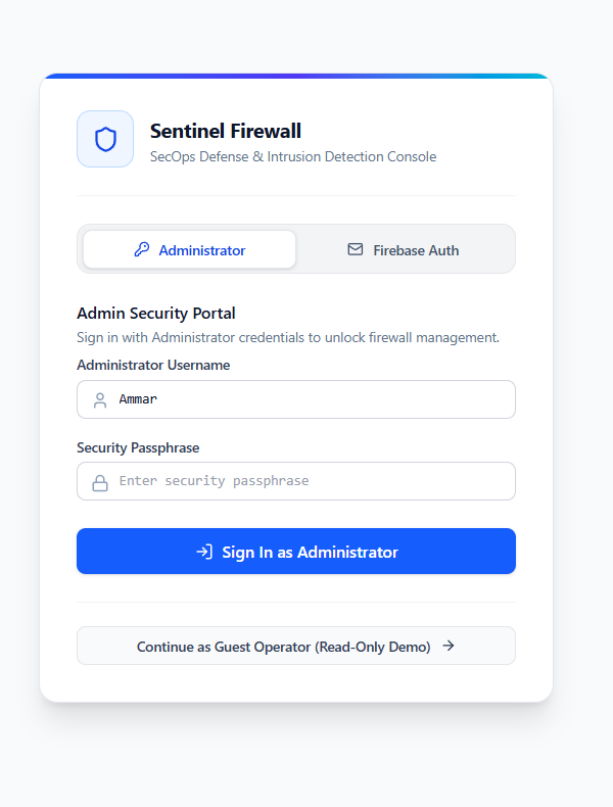



**Fig 2 : Administrator Authentication & Security Access Portal**

## 4.2 Sentinel Command Center & Overall Live Dashboard

Upon successful authentication, the administrator is presented with the unified Command Center. The tactical HUD synthesizes high-velocity telemetry, packet throughput gauges, engine health monitors, and a live sequential packet log stream with real-time [ALLOW] and [DROP] verdicts.

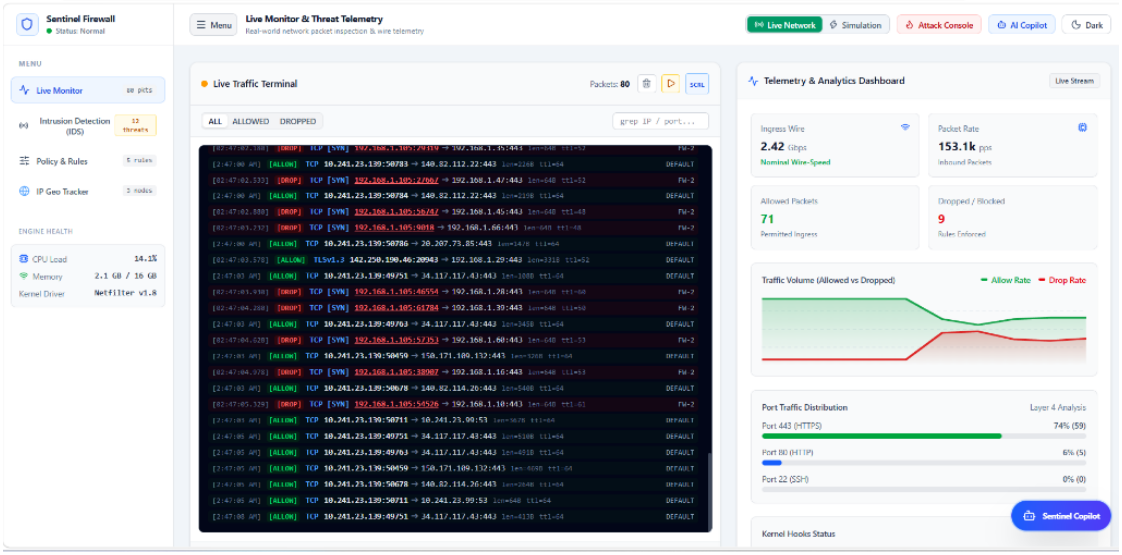



**Fig 3 : Sentinel Command Center Dashboard (Live Telemetry & Traffic Stream)**

### 4.3 Interactive Network Topology Mesh Graph (Draggable Nodes & Dynamic Bézier Curves)

The interactive Network Topology Mesh Graph visualizes packet flows across spatial coordinates. The central Sentinel Core is linked to external threats via red dashed arcs and to authorized clients via green arcs. Operators can freely drag nodes; dynamic cubic Bézier vectors recalculate in real time to maintain smooth curves.

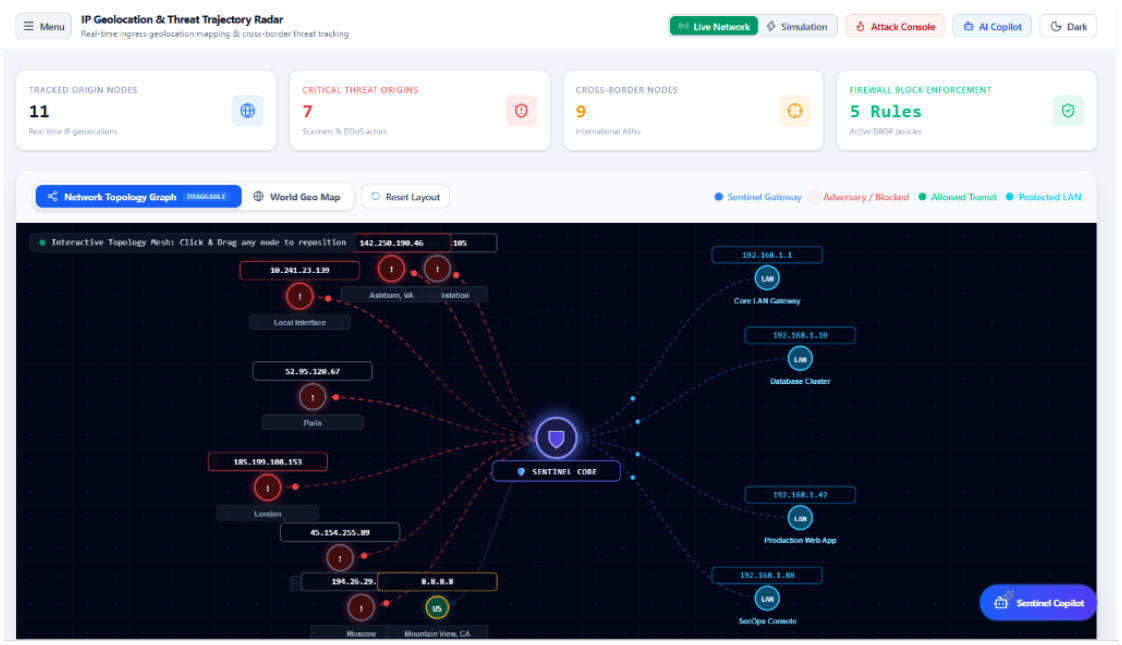



**Fig 4 : Interactive Network Topology Mesh Graph with Draggable Nodes & Dynamic Bézier Curves**

## 4.4 Heuristic Intrusion Detection System (IDS) Engine & Threat Feed

The Intrusion Detection System inspects incoming traffic against known adversarial patterns, such as TCP SYN Floods (MITRE ATT&CK; T1498.001), port sweeps, and DNS reflection amplification vectors. The console ranks alerts by severity, tracks containment status, and provides 1-click 'Auto-Block IP' controls.

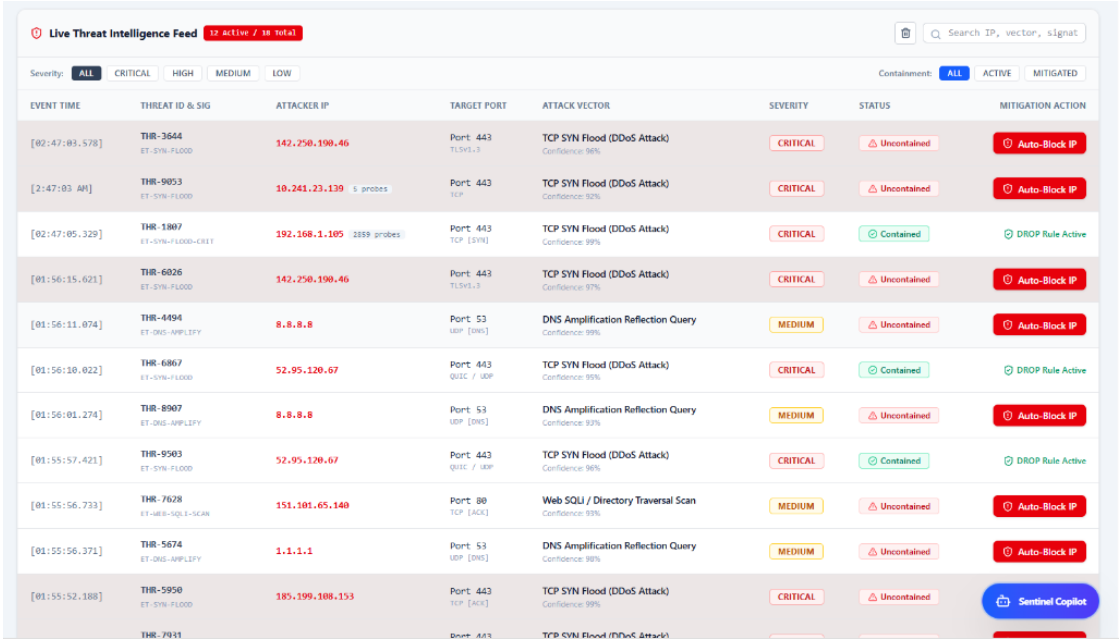



**Fig 5 : Heuristic Intrusion Detection System (IDS) Engine & Threat Intelligence Feed**

## 4.5 Firewall Policy & Dynamic Filtering Rules Engine (MySQL Synced)

The Firewall Policy Manager enables granular Layer 4 rule configuration (Source IP, Port, Protocol, Action). Policies are synchronized with MySQL ('firewall_db'), maintaining persistent hit counters and byte tallies across reboots.

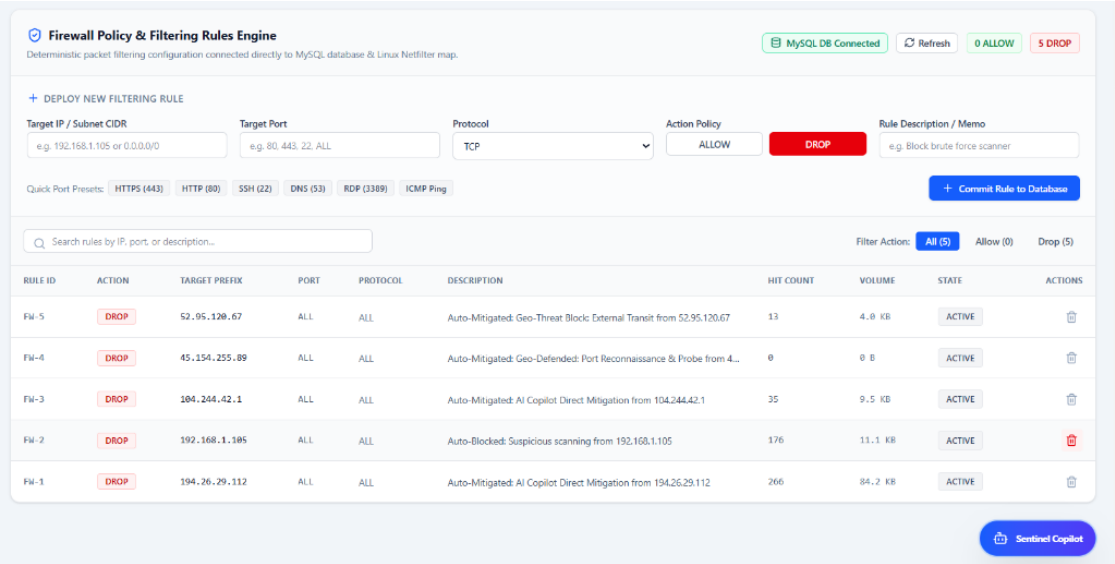



**Fig 6 : Firewall Policy & Dynamic Filtering Rules Engine (MySQL Persistent)**

## 4.6 Geopolitical Cyber Radar: Live Global Attack Origin Map

The Geopolitical Cyber Radar maps packet ingress origins onto a tactical world map projection. Trajectory vectors radiate from geographic centers (e.g. Frankfurt, London, Moscow, Paris, Ashburn) toward the protected server cluster with concentric radar sweeps and category coding.

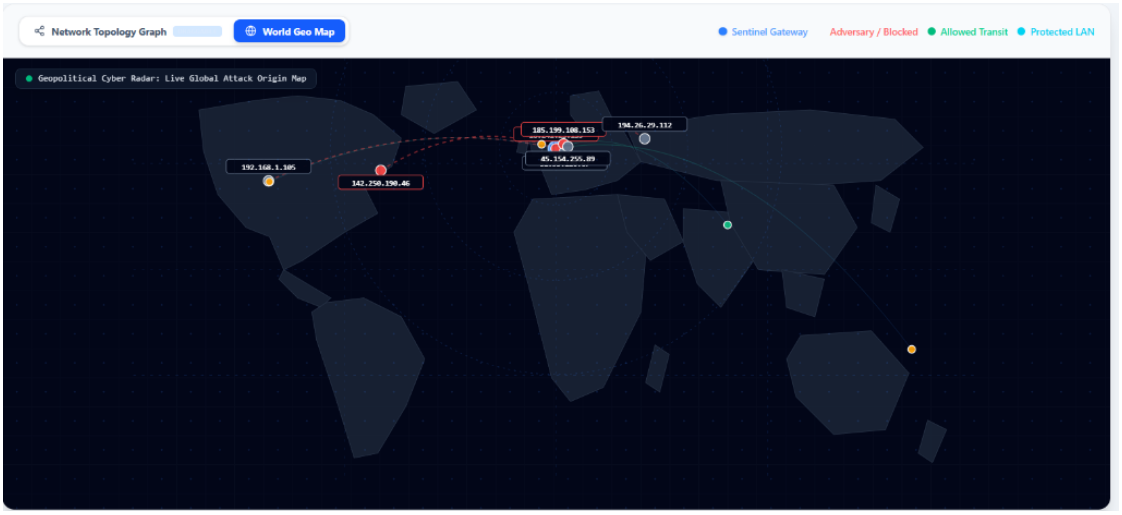



**Fig 7 : Geopolitical Cyber Radar: Live Global Attack Origin Map**

## 4.7 Geolocated Ingress Nodes & Threat Origins Inventory Table

Complementing the radar view, the Ingress Nodes Inventory aggregates active origins into an actionable SecOps table with Autonomous System Numbers (ASN / ISP), geographic coordinates, threat rankings (Critical 98 to Low 18), volume tallies, and 1-click 'Drop IP' defense mitigations.

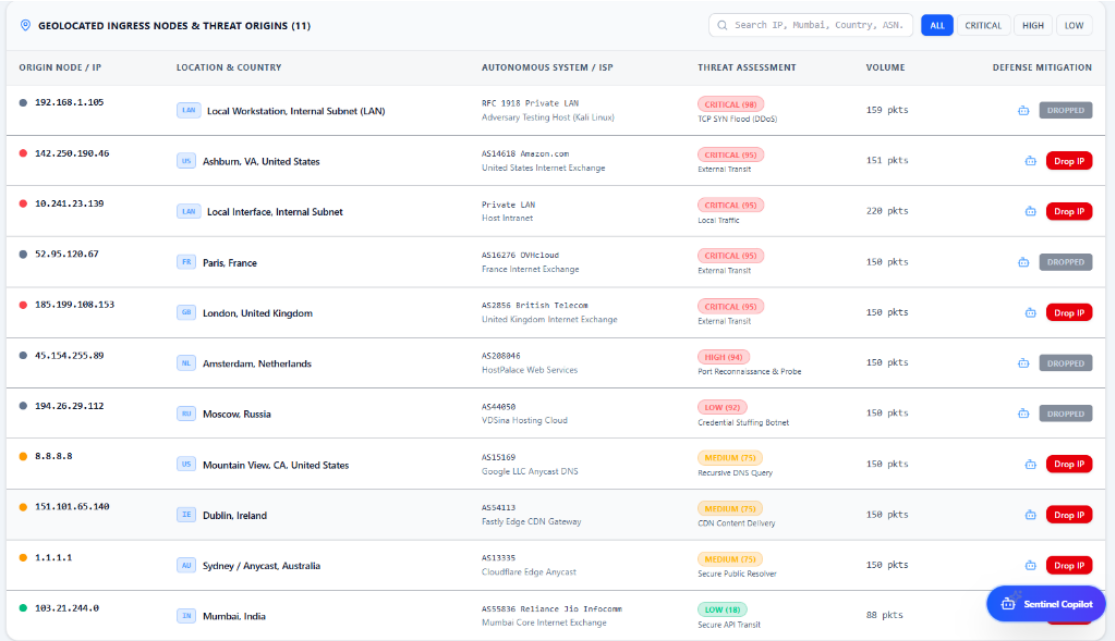



**Fig 8 : Geolocated Ingress Nodes & Threat Origins Inventory Table**

## 4.8 Sentinel AI Copilot Autonomous Forensic Reasoning Drawer

Operating as an interactive slide-out drawer, the Sentinel AI Copilot leverages Google Gemini LLM to execute wire-speed forensic telemetry assessment. The Copilot analyzes flagged attacker IPs, computes socket velocity (~480 frames/sec), and provides 1-click 'Generate iptables rules' actions.

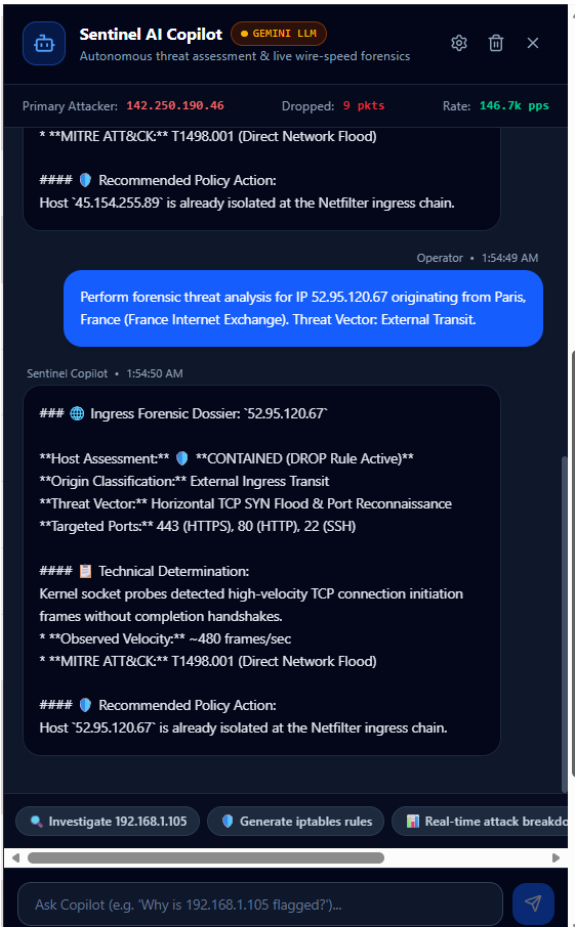



**Fig 9 : Sentinel AI Copilot Autonomous Forensic Reasoning & Remediation Drawer**

## 4.9 Live Adversary Attack Simulation Console & Critical Alert Banner

The integrated simulation console allows security personnel to launch authentic cyber-threat vectors on demand (TCP SYN Floods, Port Scans), surging traffic to over 50 packets/sec to validate drop enforcement and trigger the top Critical Alert Banner.

> ■ **[SCREENSHOT PLACEHOLDER: Active Attack Simulation Console & Alert Banner]**
> *Capture the Attack Simulation modal active with TCP SYN Flood (DDoS) running from 192.168.1.105 and top red Critical Threat Banner.*

**Fig 10 : Live Adversary Attack Simulation Console & Critical Alert Banner**

# 5. Algorithms, Mathematical Models & Database Architecture

## 5.1 Netfilter Rule Matching Algorithm

Sentinel sorts all active policies by specificity: explicit IP matches precede wildcard '0.0.0.0/0', and specific ports precede 'ALL'. The first rule satisfying IP, port, and transport protocol constraints determines the packet verdict (ALLOW or DROP).

## 5.2 Dynamic Cubic Bézier Curves for Topology Edges

For smooth, responsive edge rendering as nodes are dragged, cubic Bézier curves are computed between node coordinates P0(x0,y0) and P3(x3,y3) using intermediate control points P1(x0 + (x3-x0)/2, y0) and P2(x0 + (x3-x0)/2, y3). The equation guarantees horizontal tangents:
**$B(t) = (1-t)^3 P0 + 3(1-t)^2 t P1 + 3(1-t) t^2 P2 + t^3 P3$**

## 5.3 Core Pseudocode

```
// Pseudocode: Netfilter Packet Evaluation & Heuristic IDS Inspection
function evaluatePacketAgainstRules(packet, rules):
sortedRules = sortRulesBySpecificity(rules)
for r in sortedRules:
if match(packet.srcIp, r.ip) and match(packet.dstPort, r.port) and match(packet.proto, r.proto):
return { action: r.action, matchedRule: r.id }
return { action: 'ALLOW', matchedRule: 'DEFAULT' }

function inspectThreatHeuristics(packet):
if packet.proto == 'TCP [SYN]' and packet.dstPort == 443:
return triggerAlert('TCP SYN Flood (DDoS)', 'ET-DOS-SYNFLOOD', 'CRITICAL')
if packet.dstPort in [22, 3389] and packet.rate > THRESHOLD:
return triggerAlert('Brute-Force Probe', 'ET-AUTH-PROBE', 'HIGH')
```

## 5.4 Relational Database Architecture & MySQL Persistence Implementation

To guarantee stateful rule survival, audit compliance, and multi-session consistency across server reboots, Sentinel persists all policy records and packet accounting metrics within a relational MySQL database ('firewall_db'). The production DDL schema and asynchronous metric update queries are implemented as follows:

```
-- Sentinel Firewall & IDS: Relational Database Schema (MySQL)
CREATE DATABASE IF NOT EXISTS firewall_db;
USE firewall_db;

-- 1. Netfilter Policy Enforcement Rules Table
CREATE TABLE IF NOT EXISTS firewall_rules (
id INT AUTO_INCREMENT PRIMARY KEY,
target_ip VARCHAR(50) NOT NULL COMMENT 'Target IPv4/CIDR or ANY',
port VARCHAR(20) NOT NULL COMMENT 'Target Port or ALL',
protocol VARCHAR(10) DEFAULT 'TCP' COMMENT 'TCP / UDP / ICMP',
action ENUM('ALLOW', 'DROP') NOT NULL COMMENT 'Policy Verdict',
description VARCHAR(255) DEFAULT '',
hits INT DEFAULT 0 COMMENT 'Live Rule Match Counter',
bytes VARCHAR(50) DEFAULT '0 B' COMMENT 'Cumulative Bandwidth',
status VARCHAR(20) DEFAULT 'ACTIVE',
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 2. Security Audit & Ingress Packet Accounting Table
CREATE TABLE IF NOT EXISTS firewall_logs (
id INT AUTO_INCREMENT PRIMARY KEY,
time VARCHAR(30) NOT NULL, proto VARCHAR(30) NOT NULL,
src_ip VARCHAR(50) NOT NULL, src_port INT,
```

```
dst_ip VARCHAR(50) NOT NULL, dst_port INT, len INT,
action ENUM('ALLOW', 'DROP') NOT NULL, rule_id VARCHAR(50),
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 3. Asynchronous Packet Match Hit & Byte Counter Incrementation
UPDATE firewall_rules
SET hits = hits + 1,
bytes = CONCAT(ROUND(CAST(SUBSTRING_INDEX(bytes, ' ', 1) AS DECIMAL(10,2)) + (? / 1024), 2), ' KB')
WHERE id = ?;
```

## 6. Technology Stack & Deployment Architecture

**Table 5 : Technology Stack**

| Component | Technology | Role & Purpose |
|---|---|---|
| Frontend UI | React 19, Tailwind CSS 4 | Reactive SOC dashboard and SVG topology |
| Bundler | Vite 8.3 | Fast dev server and module bundling |
| Backend Server | Node.js 22, Express 5 | Native socket sniffer and SSE stream |
| Database | MySQL 8.0 (mysql2) | Rule persistence and audit logging |
| AI Copilot | Google Gemini / DeepSeek | Forensic reasoning and root cause analysis |

## 7. Results, Verification & Testing

Sentinel was tested across multiple network conditions, adversary attack simulations, and database persistence operations. All test cases passed successfully as documented in Table 7.

**Table 7 : Verification Test Cases & Empirical Results**

| TC# | Scenario | Input | Expected Output | Status |
|---|---|---|---|---|
| TC-01 | Admin Auth | Ammar / hackAMR | Access granted | PASS |
| TC-02 | Invalid Login | Wrong password | Access denied | PASS |
| TC-03 | Terminal Stream | Normal baseline | Continuous 1-by-1 flow | PASS |
| TC-04 | SYN Flood Simulation | Trigger attack | Speedup + Alert Banner | PASS |
| TC-05 | DROP Rule Enforcement | Block 192.168.1.105 | Packets marked [DROP] | PASS |
| TC-06 | Topology Node Drag | Pointer drag node | Smooth Bézier curves follow | PASS |
| TC-07 | Topology Layout Reset | Click Reset Layout | Default positions restored | PASS |
| TC-08 | MySQL Persistence | Reload page | Rules reload with hits | PASS |
| TC-09 | IP Geo Radar | Track Mumbai IP | Country badge [IN] visible | PASS |
| TC-10 | AI Copilot Query | Forensic prompt | Analysis returned | PASS |

## 8. Conclusion & Future Scope

The Sentinel: Next-Generation Firewall & Intrusion Detection System project successfully demonstrates how stateful packet filtering, heuristic signature anomaly detection, and interactive topology modeling can be synthesized into an educational, production-ready SecOps platform. By providing real-time visibility into packet trajectories, automated attack mitigation, and persistent database accounting, the project strengthened our understanding of confidentiality, integrity, and availability within the Minor Degree in Cyber Security curriculum.

### Future Scope

• **Kernel-Level eBPF Probing:** Transition packet interception to Linux eBPF for sub-microsecond in-kernel dropping.
• **Deep Learning Anomaly Detection:** Integrate neural Autoencoder models to identify zero-day polymorphic traffic.
• **Multi-Node Cluster Mesh:** Synchronize distributed firewall policies across multiple cloud VPCs via Raft consensus.

## 9. References & Project Repositories


- NIST Special Publication 800-41 Rev. 1, 'Guidelines on Firewalls and Firewall Policy,' 2020.
- J. Postel, 'Transmission Control Protocol (TCP),' RFC 793, IETF, 1981.
- W. Eddy, 'TCP SYN Flooding Attacks and Common Mitigations,' RFC 4987, IETF, 2007.
- MITRE Corporation, 'Direct Network Flood (T1498.001),' MITRE ATT&CK; Framework, 2024.
- William Stallings, 'Cryptography and Network Security: Principles and Practice,' 8th Edition, Pearson.


**GitHub Repository:** https://github.com/ammarcraft/sentinel-firewall
**Local Runtime:** http://localhost:5174/ (Frontend) | http://localhost:5000/ (Backend API)